

Using life cycle assessment to evaluate some environmental impacts of gold production

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ABSTRACT

The environmental profile of gold production with regards to embodied energy, greenhouse gas emissions, embodied water and solid waste burden has been assessed using life cycle assessment methodology. Both refractory and non-refractory ores were considered, with cyanidation extraction followed by carbon in pulp (CIP) recovery assumed for non-refractory ore processing. Flotation and pressure oxidation were included prior to cyanidation for processing refractory ores. For a base case ore grade of 3.5 g Au/t ore, the life cycle-based environmental footprint of gold production was estimated to be approximately 200,000 GJ/t Au, 18,000 t CO₂e/t Au, 260,000 t water/t Au and 1,270,000 t waste solids/t Au for non-refractory ore. The embodied energy and greenhouse gas footprints were approximately 50% higher with refractory ore due to the additional material and energy inputs and gold and silver losses associated with the additional processing steps required with this ore. The solid waste burden was based on an assumed strip ratio of 3 t waste rock/t ore, but this ratio varies considerably between mines, significantly influencing the estimated value of this impact. The environmental footprint of gold production (per tonne of gold produced) was shown to be several orders of magnitude greater than that for a number of other metals, largely due to the low grades of ore used for the production of gold compared to other metals.

The mining and comminution stages made the greatest contribution to the greenhouse gas footprint of gold production, with electricity being the major factor, and being responsible for just over half of the greenhouse gas footprint. This result emphasises the need to focus on these stages in any endeavours to reduce the embodied energy and greenhouse gas footprints of gold production. However, the significance of the contribution of the mining and comminution stages to the environmental footprint also means that falling gold ore grades will have a major impact on the environmental profile, and this issue is examined in the paper. Some technological developments in gold ore processing that have the potential to reduce the environmental footprint of gold production are also discussed.

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1. Introduction

From ancient times to present day, gold has been highly valued by society and has been the most highly sought after precious metal, often having been used for a variety of different purposes such as money, to back currency, in jewellery-making, and in dentistry. More recently, gold has been used in a wide range of electronic applications due to its high conductivity. Some of the special attributes of gold that make it highly prized by society include:

- it is one of only a few common metals which are coloured, with its bright yellow colour and shiny lustre giving it a perception of beauty;
- it is one of only a handful of metals that are so unreactive they do not tarnish;
- it is malleable and easily worked (e.g. jewellery);
- it is a good conductor of heat and electricity;
- it is comparatively rare.

Gold differs from most other metals in that the majority of the metal that has ever been mined is still in use. It has been estimated that about 15% of all gold ever mined was used in dissipative industrial applications or is unaccounted for or unrecoverable, leaving about 85% (i.e. between 133,000 and 153,000 t – see later) still in use and available for recycling (Muller and Frimmel, 2010;

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George, 2008). Today, just over 50% of gold production is used in jewellery, amounting to about 2000 t annually worldwide, as shown in Fig. 1.

Sustainability concerns have seen the gold industry, like other metal production industries and industrial sectors, come under increased pressure to reduce its environmental footprint over the various processing stages in its supply chain from gold ore mining through to gold refining. To this end, this paper describes a life cycle assessment study carried out to provide indicative estimates of the environmental profile of gold production in terms of energy, greenhouse gases, water and solid wastes, some of the issues affecting the profile, and some technological developments that have the potential to reduce it. The environmental profile considered was not meant to be comprehensive, but rather a selected number of environmental impacts were chosen for detailed investigation. Cyanide toxicity is a significant environmental issue in gold ore processing and the International Cyanide Management Code (www.cyanidecode.org) established by the gold industry requires management of cyanide in the effluent streams from gold processing, and a number of cyanide destruction processes have been developed (Harcus, 2011) as well as cyanide recovery technologies for copper containing gold ores (Dai et al., 2012). However, while briefly referred to in the paper, toxicity was not one of the environmental impacts assessed as it was outside the scope of the study.

2. Gold ores and mineralogy

Gold is found in two major types of deposits. Lode (or vein) deposits are deposits where gold is found embedded in cracks and veins in rocks. The second type of gold deposit is placer (or alluvial) deposits which are formed by moving water that has eroded gold out of lode deposits and deposited it in sand, crevices and stream beds. Copper and iron are the most common impurities in gold ores. Native gold is by far the most common form of gold in ores, with a gold content of 90% or more and frequently accompanied by silver. After native gold, the gold–silver tellurides are the most common gold minerals. Apart from the discrete gold minerals, gold occurs as a trace element in several common sulphides and sulpharsenide minerals (Vaughan, 2004). From a metallurgical perspective, gold ores can be broadly subdivided into free-milling (or non-refractory) and refractory types. The former are relatively easy to treat by conventional technology (crushing, grinding, density separation and cyanidation), while refractory ores require more complex processing (e.g. additional steps of flotation, roasting, bacterial or pressure oxidation prior to cyanidation) for gold recovery. Refractory ores are generally regarded as those where the gold is in some way locked¹ in the sulphide fraction. Approximately 10% of world gold production is from refractory ores (Yen et al., 2008). However, in recent years, as high grade and non-refractory gold ore deposits have become progressively depleted, the ability to recover gold from refractory low grade ores has become more important. In some cases, gold recovery is the primary reason for mining the ore, but in other cases gold is essentially a by-product of recovering one or more other metals (mainly copper). Low grade ores are commonly considered to contain 0.5–1.5 g/t Au.

3. Gold resources, production and demand

World gold reserves are estimated currently to be in the order of 51,000 t (USGS, 2011), and this has not changed significantly over the last two decades according to Mudd (2007a, b). Australia has

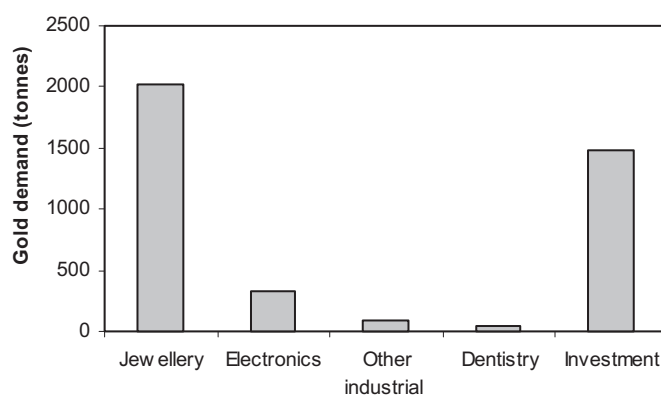


Fig. 1. World gold demand (World Gold Council, 2011).

the highest proportion of these reserves (14%) followed by South Africa (12%) and Russia (10%). However, the grade (i.e. metal content) of gold ores has been progressively falling globally over the last century (Mudd, 2007c; Muller and Frimmel, 2010), and the current world mean ore grade is in the order of 3–4 g/t Au. Muller and Frimmel (2010) suggest that based on the decrease in world mean ore grade over the last four decades, the world mean gold ore grade could fall to about 1 g/t Au in 2050. Computers, mobile phones and other electronic devices represent a large resource of potentially recoverable materials, including gold. Electronic scrap (E-waste) has been reported to contain, on average, in the order of 10.4 g/t of gold (USGS, 2001). In the case of mobile phones (excluding batteries) the gold content is in the order of 300–350 g/t while for computer circuit boards it is 200–250 g/t (Hagelucken and Corti, 2010; UNEP, 2009). However, the “mineralogy” of the scrap is much different to ore. Estimates of the historic cumulative production of gold ranges from 157,000 t to 180,000 t (Muller and Frimmel, 2010). Annual world gold mine production in 2010 was in the order of 2550 t, with about 14% of this produced in China followed by Australia (10%) and the United States (9%) (USGS, 2011; World Gold Council, 2011). World gold production appears to have peaked at 2600 t in 2001 (Denham, 2009). A further 1645 t of recycled gold was added to supply in 2010, with world identifiable gold demand in 2010 being 3970 t (World Gold Council, 2011).

4. Processing routes

The various processing routes for gold extraction from ores have been reviewed by LaBrooy et al. (1994) and Marsden (2006). Marsden and House (2006) list the various factors affecting processing route selection for metallic ores, two of which have a direct impact on gold extraction chemistry and process selection – mineralogy (as noted above) and metallurgy.

4.1. Mining

Gold production mainly comes from hard rock ore deposits, where the gold is encased in rock, rather than as particles in loose sediment.² Both open-pit (surface) and underground mining methods are used, with open-pit mining being the favoured mining method in recent years. Most of the gold production in Australia and the United States comes from open-pit mines. The advantage of

² Deposits with liberated gold in sediment or alluvial form are referred to as placer deposits, and although significant in the past, placer deposits have little economic impact today.

¹ Locked in the sense that cyanide solution is unable to access the gold.

open-pit, as opposed to underground, mining is that it is usually easier, cheaper and quicker to bring into production. Open-pit deposits often have a relatively short life-span (on average, four to five years), after which it may become necessary to move to underground mining techniques to access deeper ores (if they are available and economic). However, the continued identification of new surface deposits often provides replacement ore reserves.

4.2. Metal production

A general processing flowsheet for the recovery of gold from its ores is shown in Fig. 2, while Table 1 outlines how the various processes in Fig. 2 apply to gold ores of different grades and mineralogy (World Gold Council, 2006). A more comprehensive flowsheet of the various processing options is given by Marsden and House (2006).

4.2.1. Extraction

Cyanide leaching is the standard method used for recovering most of the gold throughout the world today (approximately 83% according to Karahan et al., 2006). The gold-bearing ore is crushed and ground to approximately 100 μ . Lime, cyanide and oxygen are then added to the ground and slurried ore. The lime raises the pH, while the oxygen and cyanide oxidise and complex the gold respectively. Leaching of high grade ores is usually carried out in tanks or vats, while heap leaching is usually applied to low grade ores. Agitated cyanide leaching accounts for about 50–55% of gold extraction (Marsden, 2006) while approximately 10% of world gold production is extracted by heap leaching (Marsden and House, 2006; Marsden, 2006). Heap leach gold recovery is typically 70%, compared to 90% in an agitated plant. In recent years, toxicity

concerns regarding the use of cyanide as a leach reagent have seen a number of alternative lixivants being proposed (Hilson and Monhemius, 2006), with thiourea and thiosulphate considered as the most realistic substitutes, with leaching rates comparable to cyanide or better (Marsden and House, 2006; Tanriverdi et al., 2005; Aylmore and Muir, 2001). Despite widespread research, thiourea and thiosulphate have not been widely adopted in the gold industry mainly due to their high consumption and/or cost, and cyanide will continue to be the only practical leach reagent in large-scale gold extraction processes while this economic advantage remains (Hilson and Monhemius, 2006).

Gravity concentration works when gold is in a free elemental state in particles large enough to allow mechanical concentration to occur. Placer mining is generally where gravity concentration has been most widely applied. Sometimes the gravity gold concentrates are cleaned up by amalgamation, whereby free gold particles are wetted by mercury and form a gold–mercury amalgam and can thus be separated from most impurities. Mercury is subsequently separated from the gold by distillation. This method was once widely used in gold processing but has gone out of favour due to the health and environmental hazards associated with mercury and the inferior performance compared with alternative processes. However, it is still used in small-scale mines, particularly in third world countries (Veiga et al., 2006).

4.2.2. Recovery

Gold is usually recovered or extracted from the cyanide solution by one of two methods: carbon adsorption or Merrill-Crowe recovery. In the more commonly used carbon in pulp (CIP) adsorption process, pellets of activated carbon are added to the leach slurry, with the gold–cyanide complex being adsorbed on the

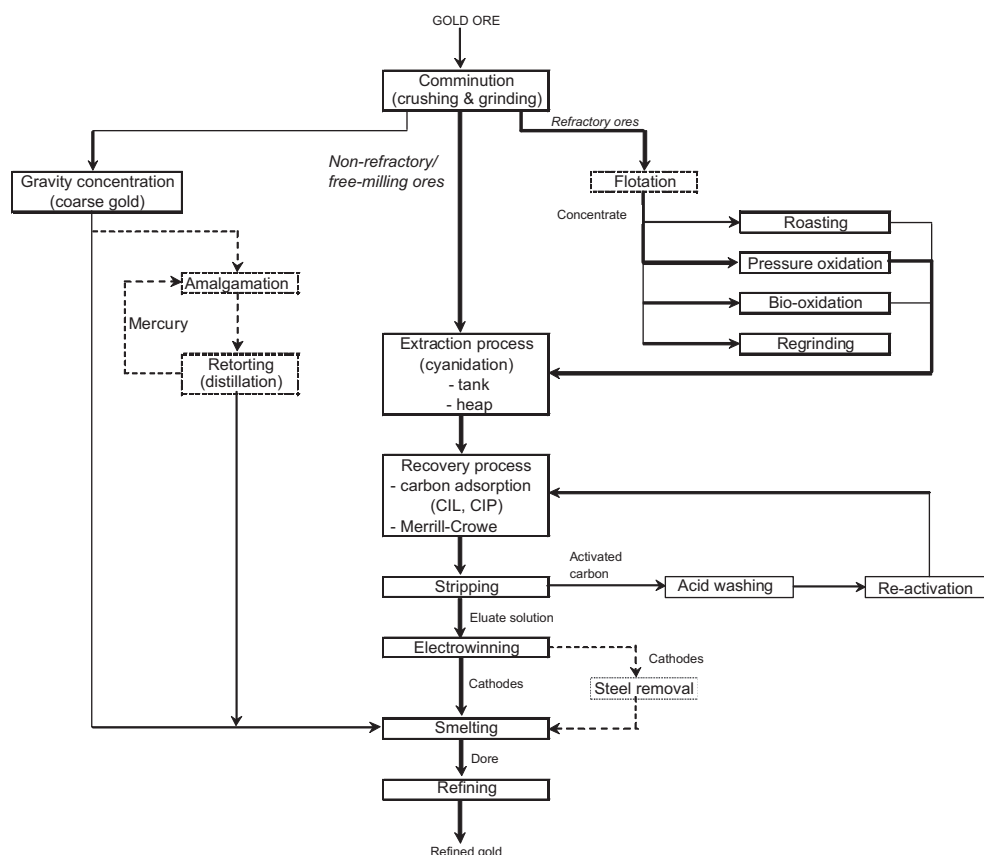


Fig. 2. General processing flowsheet for gold ores.

Table 1
Gold ore treatment processes.

	Oxide	Sulphide	Mixed sulphides	Refractory
High grade	Carbon in leach (CIL)	Carbon in leach (CIL)	Flotation to concentrate then leach at mine	Roaster
	Heap leach	Carbon in pulp (CIP) if ore a little refractory	Flotation to concentrate then leach at mine	Bio-oxidation
	Merrill-Crowe (high Ag)	Merrill-Crowe (high Ag)	Flotation to concentrate	Autoclave
Low grade	Heap leach	Gravity concentration then CIL	May not be viable	Fine grind

carbon. The loaded carbon is then separated from the pulp by screening, with the stripped solution being recycled. The carbon pellets are then transferred to a stripping or elution circuit where the carbon is washed by a hot caustic cyanide solution to reverse the adsorption process (desorption) and strip the carbon of gold. Gold is then removed from the solution by electrowinning onto steel wool. The steel wool is either dissolved away by hydrochloric acid, leaving a residue which is smelted into gold doré³ bars in crucible furnaces to produce unrefined bullion, or alternatively removed in the slag from the furnace. The stripped or desorbed (barren) carbon is regenerated by heating prior to being reused. The carbon in leach (CIL) variation of this process integrates leaching and carbon in pulp adsorption into a single unit process operation, and is frequently used when treating ores which contain carbonaceous materials which adsorb gold prematurely from the leach liquor (this phenomenon is commonly referred to as “preg-robbing”). The carbon added in CIL is more active than native carbon, so the gold will be preferentially adsorbed. The CIP and CIL processes account for approximately 42% of worldwide gold production (Marsden, 2006). In the Merrill-Crowe process, the gold is recovered from the cyanide solution by zinc precipitation (gold precipitates out of solution by a simple replacement reaction with zinc powder) and solid/liquid separation is then required.

4.2.3. Refining

There are a number of gold refining processes. In the chlorination (Miller) process, chlorine is introduced to melted bullion in a crucible furnace. The gas reacts with silver and any remaining base metals to form chlorides. At the operating temperature of the Miller process, any zinc and lead chlorides are volatile, while silver and copper chlorides accumulate on the surface of the molten bullion as a slag which is removed. The molten, refined gold is cast into bars. The electrolytic (Wohlwill) process involves dissolving gold from the bullion (anode) in a chloride solution and redepositing the gold on a pure gold or titanium cathode. The cathodes are melted and cast. The anodes for the electrolytic process are commonly made from gold that has already been through the chlorination process. The chlorination process produces gold of about 99.9% purity (which is suitable for monetary bullion), whilst the electrolytic process produces gold of 99.99% purity. Although refined gold from the chlorination process is in a marketable form, according to Habashi (1997) high purity gold (99.99%) is today used almost exclusively for both industrial and investment purposes. Therefore both the chlorination and electrolytic refining processes were included in the LCA study described here. The slag is further treated and refined to recover the silver (Pickles, 1995; Trainor, 1993).

4.2.4. Refractory ores

When gold is finely disseminated in a sulphide host mineral, the ore often cannot be ground down fine enough to expose the gold particles for direct contact with cyanide solution. In this case the ore is pretreated, the objective of which is to remove enough of the

sulphide by oxidation (thereby converting the ore to oxide) so that at least a small portion of all gold particles are directly exposed. Processes used for pretreatment all involve oxidation of sulphur and include bio-oxidation, pressure oxidation (autoclaving) and roasting. Due to environmental problems and limited efficiency, roasting is becoming less attractive, with hydrometallurgical oxidation being the preferred method. Pressure oxidation is widely applied, using sulphuric acid at elevated temperatures. The amount of refractory ore to be pretreated is greatly reduced by first producing a finely ground concentrate. The degree of oxidation required depends on the ore mineralogy and the type of oxidation process used. Systems that treat low sulphur materials (i.e. approximately <2% S) and/or materials that consume acid may require large addition of fresh acid to maintain the desired acid concentration. On the other hand, high sulphur materials (i.e. >3% S) generate sufficient acid by their decomposition and the need for supplemental acid is usually limited to any feed preparation requirements (Marsden and House, 2006).

5. Life cycle assessment

As noted earlier, the mining, mineral processing and metal production sector is coming under increased pressure to reduce its energy and water consumption along with its greenhouse gas emissions and improve the overall sustainability of its operations. Life cycle assessment (LCA) is an internationally standardised methodology that has been developed to account for the environmental impacts over a product's life cycle from raw material acquisition to the production, use and disposal of the material or product. LCA methodology is increasingly being used to assess the environmental sustainability of metal production processes (Norgate et al., 2007; Norgate and Jahanshahi, 2011), but does not appear to have been applied to any significant extent to gold production.

Like many metals, there is a range of processing routes for gold ores depending on ore mineralogy, grade, etc., as outlined above. For this study the selected main processing routes are cyanide leaching, followed by CIP recovery, electrowinning and refining (chlorination and electrowinning) for non-refractory ores. In the case of refractory ores the additional steps of flotation and pressure oxidation leaching are included prior to cyanide leaching. These two processing routes are emphasised by bold arrows in Fig. 2.

5.1. LCA assumptions

The following assumptions were made in carrying out the LCA study:

- open-pit mining
- strip ratio 3 t waste rock/t ore (Mudd, 2007a)
- gold is main metal product, not a by-product from recovery of other metals
- two ores – refractory and non-refractory
- ore grade
 - 3.5 g Au/t (base case) – range 0.5–5 g/t considered

³ A bar of semi-refined gold (i.e. bullion) with a typical gold content of 90%.

- 0.6 g Ag/t (base case) – Ag/Au ratio maintained when gold grade varied
- extraction processes
 - cyanidation (tank and heap) – non-refractory ore
 - flotation/pressure oxidation⁴/cyanidation (tank) – refractory ore
- recovery processes
 - carbon in pulp (CIP) leaching process
 - electrowinning
 - smelting
- refining
 - chlorination and electrorefining
- gold recoveries
 - flotation 91% (refractory)
 - pressure oxidation 90% (refractory)
 - cyanidation 95%
 - CIP 95%
 - EW and refining 99.9%
- silver recovery
 - to doré 60%
 - refining 99.9%
- doré (bullion) contains 90% gold and 10% silver.

The LCA was carried out using the SimaPro (version 7.3) software program.

5.2. Allocation

As it has been assumed that the gold ore also contains silver, a large proportion of which is recovered into the doré, it is necessary to allocate some of the environmental impacts of processing this ore to the silver co-product produced as well as the main gold product. The two most common methods used in LCA for co-product allocation are based on mass and economic value of the various co-products. As the price of gold is typically 40–50 times that of silver, allocation on the latter basis will attribute most of the environmental impacts to the gold. In an LCA of gold production, [Rio Tinto \(2006\)](#) used mass-based allocation for the concentrator and revenue-based allocation for the refinery. In this study both methods of allocation were used for comparative purposes, with the gold/silver price ratio assumed to be 45.

5.3. Inventory table

Table 2 lists the inventory data used in the LCA study of gold production. These data were collected from numerous sources, including published papers and reports as well as company websites, with mean values given in Table 2. There was insufficient detail in the reported inventory data to break down the “Miscellaneous” component further, but this may include other operations such as drilling, dewatering, ventilation, lighting, etc., depending on the facility. Inventory data for raw (or fresh) water consumed in gold production are given by [Norgate and Lovel \(2006\)](#).

6. Results

The embodied energy and greenhouse gas results from the LCA study are given in Table 3 for both refractory and non-refractory ores in various units and for both mass and economic co-product allocation. The embodied water and solid waste burden impacts are also included in this table, but there was insufficient detail in the inventory data to break these impacts down further for

Table 2
Inventory data used in study.

Mining		Diesel fuel	5.3 kg/t ore
		Explosives	1.7 kg/t ore
		Waste rock	3 t/t ore
Comminution		Electricity	17.7 kWh/t ore
		Steel balls	0.71 kg/t ore
Extraction and recovery	Flotation ^a	Electricity	3 kWh/t ore
		Reagents ^b	154 g/t ore
		Concentrate	0.1 t/t ore
		Tailings	0.9 t/t ore
	Pressure oxidation ^a	Electricity ^c	121 kWh/t conc
		Oxygen	0.23 t/t conc
		Fuel ^d	68 MJ/t conc
			1.3 kg/t conc
	Cyanidation	Sulphuric acid	98 g/t conc
		Electricity ^e	1.4 kWh/t ore
		Lime	2.2 kg/t ore
		Sodium cyanide	0.64 kg/t ore
		Tailings (non-refractory ore)	1 t/t ore
		Tailings (refractory ore)	0.1 t/t ore
	CIP	Electricity	5.8 kWh/t ore
		Carbon	24 g/t ore
		Sodium cyanide	0 kg/t ore ^f
		Sodium hydroxide	0.12 kg/t ore
		Hydrochloric acid	83 g/t ore
	Electrowinning	Electricity	3100 kWh/t Au
		Steel wool cathodes	0.25 t/t Au
		Hydrochloric acid	0.49 t/t Au
	Smelting	Natural gas	0.35 GJ/t Au
			6.6 kg/t Au
	Miscellaneous	Fuel ^g	0.3 kg/t ore
		Electricity	8.6 kWh/t ore
Gold refining	Chlorination process	Electricity	480 kWh/t doré
		Chlorine	0.07 t/t doré
	Electrolytic process	Electricity	325 kWh/t gold
Silver refining	Electrolytic process	Electricity	630 kWh/t silver

^a Refractory ore.

^b Dependent on ore mineralogy.

^c Includes oxygen supply.

^d Assumed to be natural gas (53 MJ/kg) – process is autothermal if feed contains 3–4% sulphur ([Linge, 1992](#)).

^e Includes air injection.

^f Included in sodium cyanide consumption for cyanidation.

^g Assumed to be fuel oil (41 MJ/kg).

Table 3
Environmental impacts of gold production.

Impact	Non-refractory ore		Refractory ore	
	Mass ^a	Economics ^b	Mass ^a	Economics ^b
Embodied energy				
GJ/t Au	199,390	221,570	303,640	337,420
GJ/oz Au	5.66	6.29	8.62	9.58
GJ/t ore milled	0.70	0.70	0.87	0.87
Greenhouse gases				
t CO ₂ e/t Au	17,560	19,520	26,840	29,820
t CO ₂ e/oz Au	0.50	0.55	0.76	0.85
kg CO ₂ e/t ore milled	61.7	61.7	77.2	77.2
Water ^c				
t/t Au	259,290	288,140	259,290	288,140
Solid waste ^c				
t/t Au	1,264,780	1,405,510	1,264,780	1,405,510

^a Co-product (Au & Ag) based on relative mass of co-products.

^b Co-product (Au & Ag) based on relative economic value (i.e. revenue) of co-products.

^c Insufficient detail in data to breakdown into refractory and non-refractory ore type.

⁴ The most common type of oxidative pretreatment ([Marsden and House, 2006](#)).

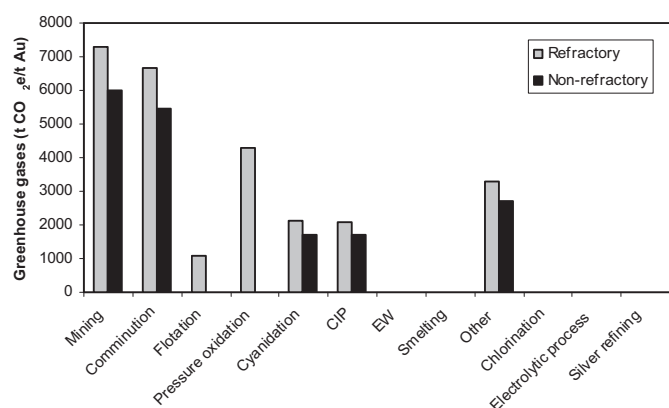


Fig. 3. Stage contributions to greenhouse gas footprint.

refractory and non-refractory ores. The greater environmental impacts with refractory ore are due to the energy and material inputs associated with the additional processing steps (flotation and pressure oxidation) required to process the ore, combined with the resulting lower overall gold and silver recovery. The environmental footprint of gold production is greater with economic co-product allocation compared with mass-based allocation as more of the environmental impacts associated with the various processing stages are assigned to the gold due to its higher economic value relative to silver, as noted earlier. Only the mass-based allocation results are shown in the following figures.

The mining and comminution stages made the greatest contribution to the greenhouse gas footprint of gold production as shown in Fig. 3. Of the various inventory inputs given in Table 2, electricity was the major contributor to life cycle-based embodied energy and GHG emissions, amounting to about 52% for non-refractory ore and about 61% for refractory ore. The contributions of the electrowinning, smelting, gold and silver refining stages are very small compared to the other stages and are not visible in Fig. 3. This is despite the appreciable electricity or fuel consumption for these stages given in Table 2, but reflects the fact that the latter inputs are expressed per tonne of gold whereas the inputs for the other stages are expressed per tonne of ore, with the low ore grade amplifying these inputs when expressed per tonne of gold.

7. Discussion

7.1. Comparison with other metals

The LCA results for gold production obtained in this study are compared with those reported for a number of other metals (Norgate and Lovel, 2006; Norgate et al., 2007) in Figs. 4–6⁵ for embodied energy, embodied water and solid waste burden respectively. These figures show that the environmental footprint of gold production (per tonne of gold produced), with regards to the environmental impacts shown, is greater than that for the other metals shown by several orders of magnitude. This observation is largely attributable to the low grade of ores used for gold production compared to ores used in the production of most other metals. Furthermore, a significant portion of the embodied energy (and associated greenhouse gas emissions) of gold production is in the mining and mineral processing stage, unlike most other metals

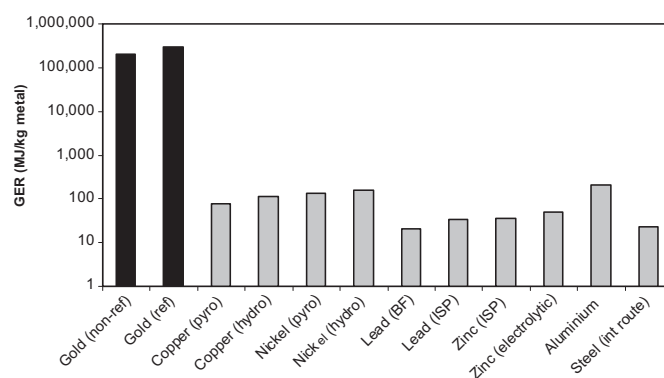


Fig. 4. Comparison of embodied energy for gold production with other metals.

where the metal extraction and refining stage is the major contributor for current world average ore grades, as shown in Fig. 7. It is this stage (mining and mineral processing) therefore that offers the greatest opportunity for reducing the energy and greenhouse gas footprint of gold production. While the specific energy footprint of gold production is significantly higher than those of other metals, when the global production of the various metals is taken into account, gold is found to have a much lower global energy and greenhouse gas footprint compared to steel and aluminium as shown in Fig. 8 (energy only shown).

7.2. Sensitivity analysis

The results of any LCA study are particularly dependent on the inventory data used and the assumptions made in the study. In this study the best available inventory data were collected from various sources as mentioned earlier, and assumptions deemed appropriate for typical gold processing operations were made, also outlined earlier. However, in order to assess the effect of these data and assumptions on the LCA results, in particular embodied energy, a sensitivity analysis was carried out by varying selected parameters by $\pm 25\%$ about their base case value.⁶ The results of this analysis are shown in Fig. 9 for refractory ore. Similar sensitivities were obtained for non-refractory ore. The results in this figure show that the embodied energy was particularly sensitive to both ore grade and overall gold recovery. As both of these parameters affect the amount of refined gold produced, it is not surprising that they have similar effects. However, changes in ore grade over the range considered are more likely in practice than similar changes in overall gold recovery. The effect of ore grade is examined in more detail in the next section. The embodied energy results were less sensitive to mining diesel consumption and comminution electricity consumption.

7.3. Effect of ore grade

The grade of metallic ores has been falling over time globally, and Mudd (2007a, b, c) has shown this to be particularly the case for gold ores. The effect of falling ore grades on the environmental footprint of gold production will be significant because of the major contribution that the mining and mineral processing stage makes to the overall footprint as noted above. Additional material will need to be handled and processed in this stage as the ore grade falls in order to produce the same amount of gold metal. The likely

⁵ Abbreviations in Figs. 4–7: pyro (pyrometallurgical), hydro (hydrometallurgical), BF (blast furnace), ISP/ISF (Imperial Smelting Process/Furnace), int route (integrated route).

⁶ In the case of overall gold recovery for refractory ores, this parameter could only by 20% above its base case value.

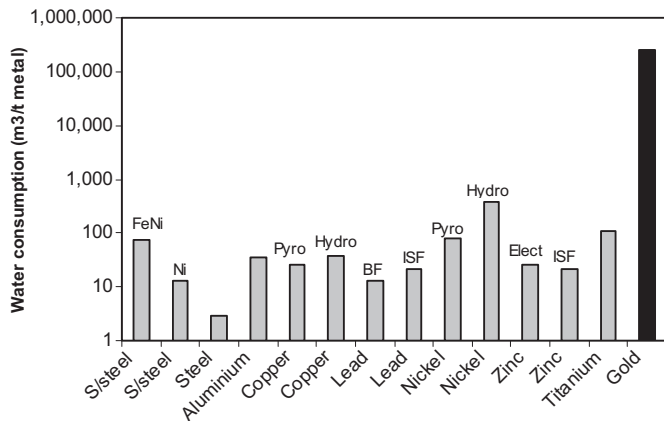


Fig. 5. Comparison of embodied water for gold production with other metals.

effects of ore grade on embodied energy, greenhouse gas emissions, embodied water and solid waste burden were obtained by altering the mined ore grade in the LCA model and are shown in Figs. 10–14 respectively. As electricity is the major contributor to embodied energy and greenhouse gas emissions as pointed out earlier, the effect of both black coal (base case) and natural gas-based electricity on the LCA results was examined, and the results are shown in Figs. 11 and 12. The greenhouse gas footprint is lower with natural gas-based electricity as it has a lower greenhouse gas intensity compared to black coal-based electricity (0.633 t CO₂e/MWh cf. 0.990 t CO₂e/MWh).

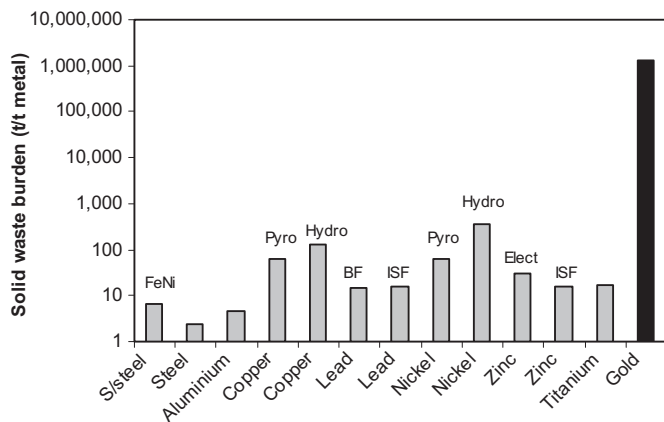


Fig. 6. Comparison of solid waste burden for gold production with other metals.

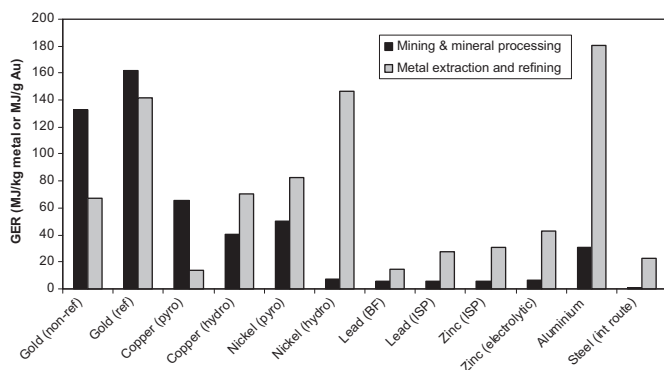


Fig. 7. Mining and milling versus extraction and refining embodied energy for various metals.

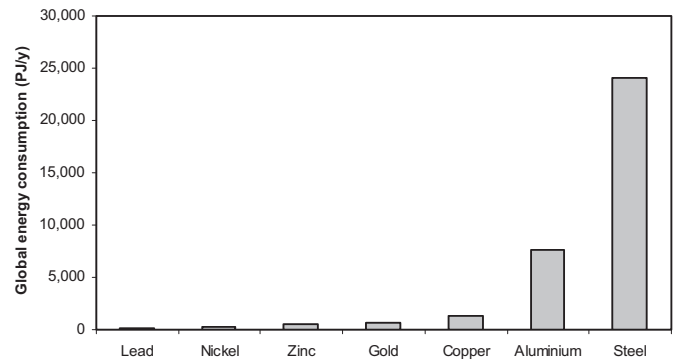


Fig. 8. Comparison of global energy consumption for metal production.

7.4. Comparison with other LCA studies

It is often difficult to compare the results of different LCA studies, even for the same metal and processing route, as issues such as ore grade and fuel mix for electricity generation are often not reported. Furthermore, it is often not clear if energy consumption and associated greenhouse gas data given in company sustainability reports are life cycle-based, i.e. whether the inefficiencies associated with electricity generation have been accounted for by converting electrical energy to primary (fuel) energy – this tends not to be the case. In addition, there appears to be a paucity of LCA studies of gold production reported in the literature. Nevertheless, some reported greenhouse gas emission data for gold production are compared with the study results in Figs. 11 and 12. The value reported by Hageluen and Meskers (2010) is taken from a life cycle inventory database so can be assumed to be life cycle-based. On the other hand, the value reported by Mudd (2007b) is largely based on data taken from company sustainability reports so can be expected not to be life cycle-based for the reason outlined above, which is probably why it is lower than the other values shown in these figures.

Water consumption for gold production for the base case ore grade of 3.5 g Au/t ore based on data reported by Mudd (2007b) is also plotted in Fig. 13 and shows good agreement with the results of the present study. The solid waste burden results shown in Fig. 14 are based on an assumed strip ratio of 3 t waste rock/t ore (Mudd, 2007a), with the waste rock produced during mining being the largest contributor to this environmental impact. However, as

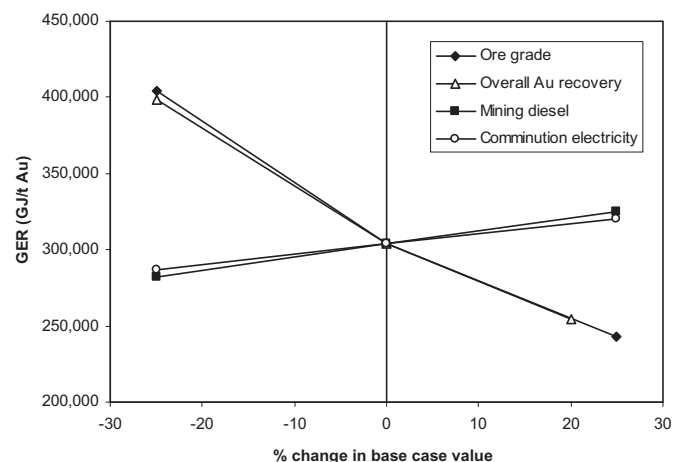


Fig. 9. Sensitivity analysis results for refractory ore.

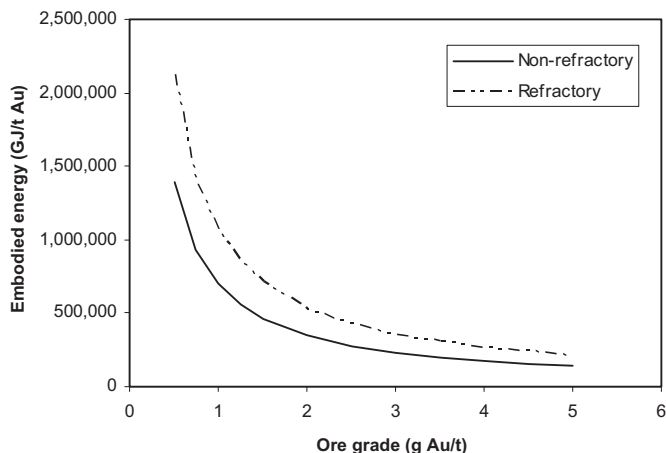


Fig. 10. Effect of ore grade on embodied energy of gold production.

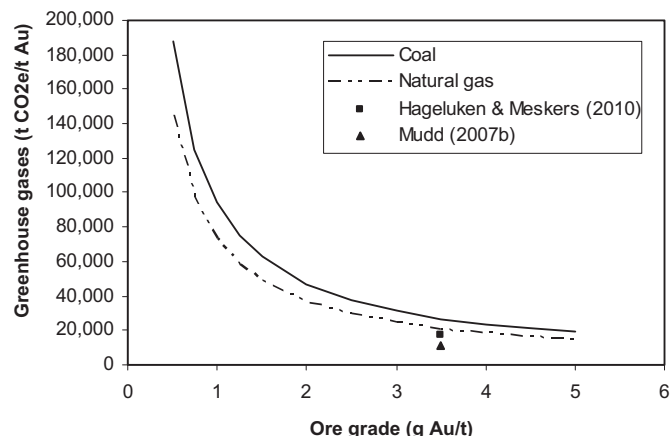


Fig. 12. Effect of ore grade on greenhouse gas emissions from gold production (refractory ore).

noted by this author (Mudd, 2007a, b), the ratio of waste rock to ore varies considerably, both between mines, mine type (open cut/underground) and over the life of a mine, can range between 2 and 10 t waste rock/t ore. Thus the solid waste burden values given in Table 3 and Fig. 14 should be viewed as indicative values only.

7.5. Opportunities to reduce the environmental footprint

As noted earlier, the greatest opportunities to reduce the environmental footprint of gold production lie in the mining and mineral processing stages. There are a number of technological developments occurring in these gold processing stages (Chadwick, 2011; Hargus, 2011; Taylor, 2010) that have the potential to reduce the environmental footprint of gold production, and some of these are described below. In the case of greenhouse gas emissions, this could be expected to come about by either reduced energy consumption, increased gold recovery, or both. Apart from examining the likely effect of oxygen injection on the greenhouse gas footprint, no estimates of the potential impacts of these technologies on the environmental footprint were made as this was beyond the scope of the present study.

7.5.1. In-place leaching

According to Norgate et al. (2010), in-situ leaching (ISL) gives embodied energy results for copper metal production comparable

with those for heap leaching. The reduced comminution requirement for heap leaching and ISL significantly reduces the energy consumption of these processes compared to conventional mineral processing requiring grinding of ores (e.g. agitated tank leaching). However, because of environmental concerns, only non-cyanide lixiviants are being considered for ISL of gold ores. The lixiviants being considered tend to break down in contact with pyrite, so the focus is on pyrite-free oxidised deposits (Taylor, 2010). As the permeability of these deposits is considered to generally be too low for ISL, permeability enhancement methods are considered to be necessary and the process is referred to as in-place leaching rather than ISL. Blasting and hydraulic fracturing are possible options for this permeability enhancement. CSIRO is conducting research into the in-place leaching of oxidised gold deposits (Roberts et al., 2009).

7.5.2. High pressure grinding rolls

High pressure grinding rolls (HPGR) technology has been widely utilised in the cement and diamond industries. More recently it has been considered for metalliferous minerals, predominantly iron, copper and gold (Van der Meer and Maphosa, 2011). McNab (2006) reported that comminution with a HPGR gave an additional 10% leach extraction for a low grade ore over conventional cone crushing. This difference was explained by the penetration of leach solution into the micro-fractures created by the HPGR. Other claimed advantages of HPGR technology are higher energy efficiency and reduced grinding media consumption (Taylor, 2010).

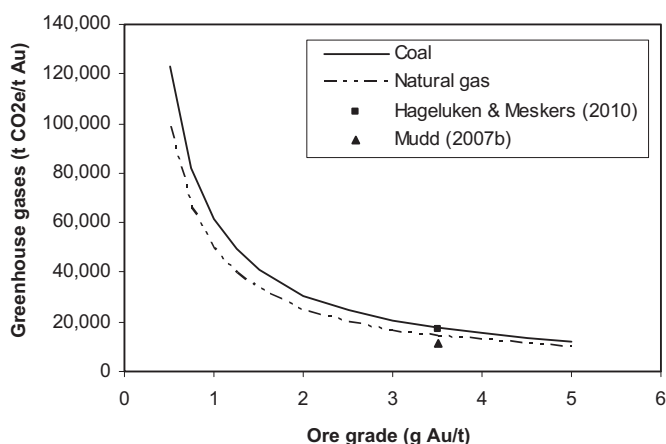


Fig. 11. Effect of ore grade on greenhouse gas emissions from gold production (non-refractory ore).

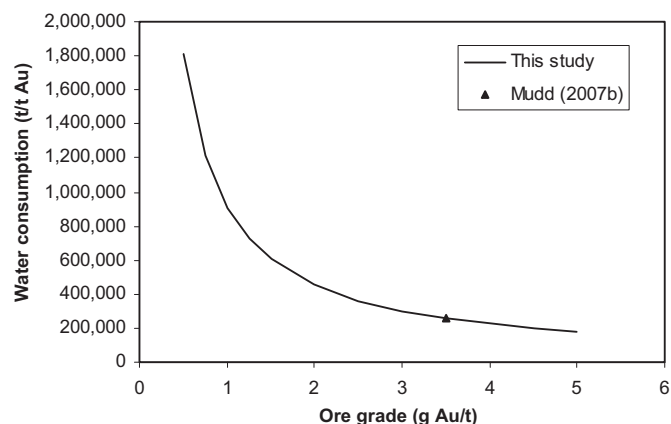


Fig. 13. Effect of ore grade on embodied water of gold production.

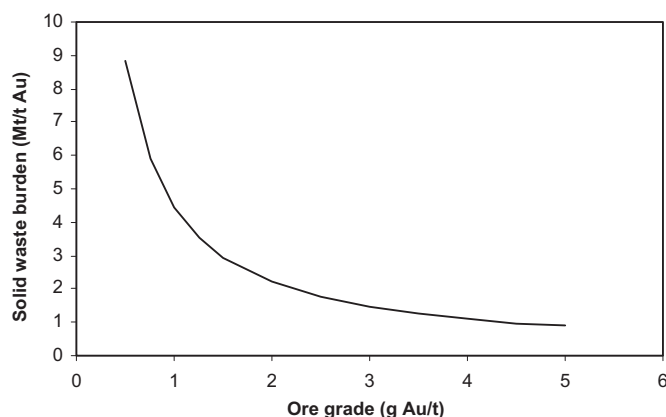


Fig. 14. Effect of ore grade on solid waste burden of gold production.

7.5.3. Flotation

While flotation has been used as the main pre-concentration process for gold ores for many years, new technologies have been developed to improve its efficiency. Magnetic aggregation has been shown to improve the efficiency of separation by improving gold recovery in the fine fraction of a low grade ore (Rivett et al., 2007). On the other hand, specially designed flash flotation circuits have been shown to increase gold recovery in a number of plants by 2.5–5% (Chadwick, 2011). Another innovative flotation technology that has been applied to gold ore processing is pneumatic flotation, which is different to conventional flotation in that the bubble–particle contact takes place outside of the cell. One of the claimed advantages of this technology is a lower power consumption (Harcus, 2011).

7.5.4. Pre-concentration using gravity concentration

Although gravity concentration is not new (see Fig. 2), recent developments in equipment type (e.g. Knelson centrifugal concentrators, inline pressure jigs (Harcus, 2011)) have seen gravity concentration used for pre-concentration of gold ores for conventional carbon adsorption or Merrill-Crowe recovery. This approach maximises the recovery of the liberated gold-bearing minerals at coarse particle sizes, thereby reducing comminution energy and yielding higher overall gold recovery, as the finer the gold becomes, the more difficult it is to recover.

7.5.5. Biological oxidation

Biological oxidation is used to treat refractory sulphide ores (see Fig. 2). The process uses a combination of three bacteria that occur naturally to break down the sulphide mineral matrix in the ore being treated, thus freeing the occluded gold for subsequent cyanidation (Chadwick, 2011). The reactors are aerated and the slurry temperature is maintained at an optimum 40–50 °C. As the oxidation rates of the sulphide minerals are exothermic, no external energy is required apart from stirring. BIOX® is the most well-known commercial process. Biological oxidation can yield higher gold recoveries than roasting and pressure oxidation for some refractory gold ores (Van Aswegen and Marais, 2001). Furthermore, as the process operates at low temperature and atmospheric pressure it can have a lower fuel requirement compared to pressure oxidation (see Table 2), although both processes can be auto-thermal if the sulphur content of the feed is high enough.

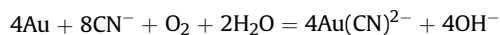
7.5.6. Microwave heating

Most metal dissolution processes are controlled by diffusion of the lixiviant from the bulk of the solution to the reaction site on the mineral of interest. Ore pretreatment by microwave heating has

been shown to result in micro-crack formation which can enhance cyanide amenability as well as grindability (Chadwick, 2011; Amankwah and Ofori-Sarpong, 2011) which could potentially increase gold recovery and reduce comminution energy requirements. Depending on the microwave energy required, this approach could lead to a decrease in the greenhouse footprint of gold production.

7.5.7. Oxygen injection

It is widely accepted that the gold cyanidation process can be represented by the following equation (Bodnaras et al., 1993):



Thus air (assumed in the base case above) or oxygen is injected into the cyanidation vessel to improve gold recovery. Increases in gold recovery rates of 1% or greater have been reported for direct injection of oxygen rather than air (Chadwick, 2011). Furthermore, when the dissolved oxygen concentration is increased, the amount of cyanide required can be decreased by as much as 25% according to Chadwick (2011). However, based on electrical power consumption of 1.1 kWh/m³ oxygen (Chadwick, 2011) for oxygen production and an oxygen consumption rate of 0.75 m³/t ore (Bodnaras et al., 1993), it was estimated using the LCA model described above that an increase of 1.3% or greater in overall gold recovery was required before the base case greenhouse gas footprint was reduced.

7.6. Gold production and sustainable development

There has been considerable debate in the literature in recent years about the sustainability of mining in general (e.g. Hilson and Murck, 2000; Whitmore, 2006; Fitzpatrick et al., 2011) and gold mining in particular (Hilson, 2002; Amankwah and Anim-Sackey, 2003; Kumah, 2006; Mudd, 2007a,b,c). It is commonly argued that the mining of mineral resources is intrinsically unsustainable as these resources are non-renewable and hence finite (Horowitz, 2006). This reasoning is fundamentally correct, as the concept of substitution, often used to respond to concerns about resource depletion, whereby if supply of a given metal becomes scarce enough its price will rise to the point where use of an alternative material becomes economic, merely extends the finite lifetime of these non-renewable resources (Richards, 2006). However, the application of the concept of sustainable development to mining is essentially about applying practices that ensures that society maximises its utilisation of the world's finite resources of metals in the most sustainable manner. Alternatively, Rankin (2011) has suggested that a more useful way of thinking about sustainability and mining is to attempt to answer the question: how can mining contribute to the transition to a sustainable society? Progress towards sustainable development goals requires improvements in both the environmental and socio-economic arenas. The latter aspect is particularly significant for artisanal and small-scale gold mining (Hilson, 2002; Amankwah and Anim-Sackey, 2003; Kumah, 2006) which has been estimated to produce roughly 10 percent of global gold production (Geoviden, 2007). The major environmental and socio-economic problems caused by gold mining in the developing world include deforestation and erosion damage; acid mine drainage; disposal of tailings into rivers; noise, dust, air and water pollution from arsenic, cyanide and mercury; social disorganisation; a loss of livelihoods and mass displacement (Kumah, 2006; Vieira, 2006). As a result of these impacts, there have been repeated calls by activist groups to have a moratorium on gold mining worldwide (Ali, 2006; Sarin, 2006). In response to this trend, the industry has shown increasing interest in environmental

and social sustainability in recent years (World Gold Council, 2012). The focus in this paper has been on the environmental aspect of sustainable development, with the mining and mineral processing stages shown to make the greatest contribution to the greenhouse gas footprint of gold production. Suggested sustainability indicators relevant for the mining industry include: energy consumption, greenhouse and pollutant emissions, water usage, solid waste, rehabilitation and land use, jobs, health and safety (Mudd, 2007a). Energy and water consumption in particular have received significant attention (e.g. Gunson et al., 2010; Kemp et al., 2010) and a number of these indicators have been used in this paper. The technological developments outlined above can be expected to improve the sustainability of gold mining, mainly through reduced energy consumption. The move towards non-cyanide leaching reagents such as thiourea and thiosulphate (Hilson and Monhemius, 2006) discussed earlier will also improve the sustainability of gold mining, as will efforts to use mercury-free processing alternatives in small-scale gold mining operations (Vieira, 2006).

Recycling and reuse are critical components of a sustainable society, and as noted earlier, about 85% of all the gold that has ever been mined is estimated to still be in use. Thus gold can be considered as a strong contributor to the sustainable use of metals, largely as a result of the high value put on gold by society due to its special attributes outlined earlier. Richards (2006) argues that the value placed on the more common metals (e.g. copper, aluminium, iron) by society should be increased to reflect the true production cost (i.e. including environmental and social impacts) plus the cost of replacement to make the use of these metals more sustainable, thereby ensuring that the “in use” stocks of these metals approach levels currently attained only by gold and other precious metals.

8. Conclusion

Gold is one of the most highly valued metals by society, however, sustainability concerns have seen the gold industry, like other metal production industries and industrial sectors, come under increased pressure to reduce its environmental footprint. In response to these pressures, the industry is endeavouring to identify opportunities to develop solutions and technologies to achieve these sustainability goals. Life cycle assessment is a useful tool that can assist in this task, and has been used in the study described here to provide indicative estimates of the environmental profile (energy, greenhouse gases, water and solid waste) of gold production from both refractory and non-refractory ores. Cyanidation extraction followed by carbon in pulp (CIP) recovery was assumed as the processing route for non-refractory ore, while flotation and pressure oxidation was included prior to cyanidation for refractory ore. Not surprisingly, the embodied energy and greenhouse gas footprints were approximately 50% higher with refractory ore due to the additional material and energy inputs and gold and silver losses associated with the additional processing steps required with this ore. The environmental footprint of gold production (per tonne of gold produced) was shown to be several orders of magnitude greater than that for a number of other metals, largely due to the low grade of ores used for the production of gold compared to other metals. However, when global metal production tonnages are accounted for, the global energy and greenhouse gas footprint of gold production is significantly less than for steel and aluminium, less than copper but greater than for zinc, lead and nickel.

The mining and comminution stages made the greatest contribution to the greenhouse gas footprint of gold production, with electricity being the major factor, and being responsible for just over half of the greenhouse gas footprint. This result emphasises

the need to focus on these stages in any endeavours to reduce the embodied energy and greenhouse gas footprints of gold production. However, the significance of the contribution of the mining and comminution stages to the environmental footprint also means that falling gold ore grades will have a major impact on the environmental profile. Technological developments in gold ore processing such as HPGR, in-place leaching, innovative flotation techniques, pre-concentration using gravity concentration, oxygen injection, biological oxidation and microwave heating have the potential to reduce the environmental footprint of gold production, but any environmental benefits from these technologies will be strongly dependent on the characteristics of the ore concerned, in particular mineralogy and grade.

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